



Xanthan gum production from sugarcane bagasse and its potential for biodegradable food packaging films

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ABSTRACT

Growing environmental concerns over plastic pollution have intensified the search for sustainable alternatives in food packaging. However, the high cost of biopolymers remains a barrier to large-scale adoption. This study addresses this gap by investigating the use of sugarcane bagasse, an abundant agro-industrial residue in Minas Gerais (Brazil), as a low-cost substrate for xanthan gum (XG) production. The objective was to evaluate the physicochemical properties of lab-produced XG and compare them to commercial XG, assessing their suitability for incorporation into biodegradable packaging films. In this sense, the developed gum was characterized for apparent and intrinsic viscosities, molecular weight, chemical/functional groups (FTIR), thermal properties (TGA and DSC), and elemental analysis (C, H, O, N and S). Fermentation of sugarcane bagasse extract (6 % w/v) with *Xanthomonas campestris* CCT 5268 at 28 °C for 120 h yielded 2.07 g/L of gum. The biopolymer exhibited pseudoplastic behavior, with an apparent viscosity of 135 mPa·s at 25 °C and 25 s⁻¹. Thermal analysis revealed degradation temperatures of 290 °C (lab-produced) and 300 °C (commercial), indicating comparable stability. The molecular weight of the lab-produced gum was 1238 kDa, surpassing the commercial sample (1018 kDa). FTIR and elemental analyses confirmed structural similarity between both samples, with minor differences attributed to the fermentation substrate. These findings demonstrate the feasibility of valorizing agro-industrial waste for the production of functional biopolymers, supporting the development of cost-effective and sustainable packaging materials.

1. Introduction

Food packaging plays a vital role in preserving product quality, ensuring protection against external contaminants, and extending its shelf life (Hadi et al., 2022; Khalid et al., 2024). In the food industry, a wide range of synthetic plastics has been employed due to their low cost, strength, flexibility, light weight, stability, and good processability, offering convenience across various packaging applications (Chen et al., 2022a; Hadi et al., 2022; Ratna et al., 2024). However, conventional plastic packaging is typically derived from petroleum-based polymers such as polyethylene (PE), polypropylene (PP), and polyethylene

terephthalate (PET), which contribute significantly to environmental waste accumulation due to their persistent use and non-biodegradable nature (Chen et al., 2022b; Elella et al., 2021; Hadi et al., 2022; Hadidi et al., 2022; Jayakumar et al., 2022; Khalid et al., 2024).

To address this issue, there is growing interest in developing innovative and sustainable packaging alternatives that combine environmental responsibility with functional performance. Thus, biodegradable polymers have emerged as promising candidates, particularly when derived from renewable and low-cost materials such as agro-industrial residues (Hadidi et al., 2022; Khalid et al., 2024). Among these, polysaccharides, proteins, and lipids obtained from natural origins,

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including agricultural wastes, plants, animals, and seaweed extracts, have gained prominence for their biodegradability and sustainable potential (Hadi et al., 2022; Khalid et al., 2024).

Agro-industrial residues represent a major opportunity for waste valorization within the principles of the bioeconomy and circular economy. Globally, the sugar industry generates approximately 279–280 million tons of sugarcane waste annually, much of which is inadequately disposed or burned, releasing greenhouse gases and affecting human health (Thongsomboon et al., 2023; Ungureanu et al., 2022). In Brazil, the world's leading sugarcane producer, an estimated harvest of 689.8 million tons for 2024/2025 season according to the Brazilian National Supply Company, generating substantial volumes of bagasse rich in carbohydrates and cellulose. Valorization of these residues into high-value bioproducts not only mitigates environmental impacts but also supports sustainable development by transforming waste into resources for energy, chemicals, and biomaterials. In this context, sugarcane bagasse stands out as a strategic feedstock for biopolymer production, offering a low-cost and abundant substrate for applications such as biodegradable packaging.

Given its functional properties, xanthan gum (XG) stands out as a promising microbial polysaccharide due to its rheological behavior, biocompatibility, and gelling properties (Elella et al., 2021; García-Ochoa et al., 2000; Murad et al., 2019; Nsengiyumva et al., 2024). Although traditionally produced from glucose or sucrose, XG can also be synthesized using agro-industrial residues such as sugarcane bagasse, cassava starch, cheese whey, and many others, which offer a cost-effective and sustainable alternative for biopolymer production (Brandão et al., 2013; Murad et al. 2019; Palaniraj and Jayaraman, 2011). These substrates provide nutrient-rich media; however, fermentation parameters, such as strain, substrate type, agitation, oxygen, temperature, and pH, strongly influence XG properties, which are critical for its potential use in packaging applications (Elella et al., 2021; García-Ochoa et al., 2000). Optimizing these variables is essential to enhance film-forming properties and expand the applicability of XG in biodegradable food packaging.

For this reason, several extraction methods have been employed, ranging from conventional solvent-based techniques to advanced non-thermal approaches for the reuse and recovery of valuable compounds from agro-industrial residues. According to Xu et al. (2025), conventional methods such as maceration, infusion, and Soxhlet extraction are widely used but often rely on organic solvents, raising environmental and safety concerns. In contrast, emerging techniques like ultrasound-assisted extraction, microwave-assisted extraction, ohmic heating, and enzyme-assisted extraction offer improved efficiency, reduced solvent usage, and better preservation of thermolabile compounds. These green technologies are increasingly applied to recover a wide range of compounds from agro-industrial residues, including bioactive substances for functional applications and polysaccharides such as pectin and xanthan gum, which are particularly relevant for biodegradable packaging. The recovery of these film-forming biopolymers supports the development of sustainable materials that integrate environmental responsibility with functional performance.

Therefore, this study aimed to produce xanthan gum from sugarcane bagasse using a simplified fermentation process, comparing its properties with those of a commercial sample. Although previous studies have demonstrated the feasibility of producing XG from agro-industrial residues at the laboratory scale (Brandão et al., 2013; Dos Santos et al., 2016; Murad et al., 2019; Nery et al., 2013; Palaniraj and Jayaraman, 2011), most have focused primarily on production yield and basic characterization. In contrast, the present study conducted a more robust physicochemical characterization of lab-produced XG, aiming to assess its functional suitability for future application in biodegradable food packaging. According to Li et al. (2025), XG-based films have shown great potential for preserving perishable food products by acting as effective barriers against moisture, oxygen, and microbial contamination, while also offering biodegradability in soil. These attributes

reinforce the relevance of evaluating XG derived from alternative substrates for sustainable packaging applications.

2. Materials and methods

2.1. Material

Xanthomonas campestris strain CCT 5268 was purchased from Fundação André Tosello (Campinas, São Paulo, Brazil). Sugarcane bagasse was obtained as a residual by-product after juice extraction in a local commercial establishment (Ouro Branco, Minas Gerais, Brazil). The material, already processed for sugarcane juice production, was collected as waste, dried in an oven at 40 °C for 24 h to remove residual moisture, and stored in a freezer until use. This residual bagasse was employed as the fermentation substrate for xanthan gum production. Xanthan gum standard 200 mesh (commercial xanthan gum) was purchased from Synth (Brazil). All other reagents used were of analytical grade.

2.2. Microorganism and growth

The methodology used for strain maintenance, xanthan gum production, and precipitation was described by Nery et al. (2008), Brandão et al. (2013), and Gomes et al. (2015) with modifications.

Xanthomonas campestris (CCT 5268) was received in 2 Falcon tubes with Yeast Malt Agar (YM agar) and they were stored in an ultra-freezer at - 80 °C with glycerol 10 % (v/v) (Sigma Aldrich) to preserve their characteristics. A small amount of the preserved culture was grown in YM liquid and used as inoculum for the fermentation process. YM Agar contains (w/v): glucose 1.0 % (Êxodo Científica); yeast extract 0.3 % (Synth); malt extract 0.3 % (Himedia); bacteriological peptone 0.5 % (Êxodo Científica); agar 2.0 % (Kasvi). The YM liquid contained the same content as the YM agar, except that agar was not added. The medium was sterilized at 121 °C for 20 min (Nery et al., 2008).

2.3. Production of xanthan gum (XG)

Three different fermentation media were prepared using sugarcane bagasse to select the appropriate substrate to produce xanthan gum using agro-industrial waste. These formulations were based on preliminary tests. Sugarcane bagasse (2 % w/v), sugarcane bagasse (2 % w/v) supplemented with sugarcane broth (2 % v/v), and sugarcane bagasse (6 % w/v) represent the amount of alternative substrate used in the production of XG.

The sugarcane bagasse extract was ground in a domestic mixer with distilled water and filtered using an organza fabric. The filtrate was used as the fermentation medium. All media were minimally supplemented to provide sources of nitrogen, phosphorus and potassium, using 0.01 % (w/v) urea (Neon) and 0.1 % (m/v) K₂HPO₄ (Lab Import). After preparing the media, they were sterilized at 121 °C for 20 min.

After media sterilization and cooling, 10 mL of inoculum was added to each Erlenmeyer flask, placed on an orbital shaker, and refrigerated at 28 ± 2 °C and 250 rpm for 120 h to produce xanthan gum. After the end of fermentation, centrifugation (3500 rpm, 1591 × g, T = 4 °C for 20 min) was carried out to separate xanthan gum from the media.

After centrifugation, ethyl alcohol (92.8 °GL) (Itaja) at a ratio of 3:1 (ethanol: fermentation broth) (v/v) was added to the supernatant to precipitate the gum for 24 h at room temperature. In sequence, the floated gum was frozen (T = -12 °C) and lyophilized in L108 equipment (Liotop, Brazil) at a temperature of -54 °C for approximately 24 h. The xanthan gum yield was determined by gravimetry and expressed as g/L.

2.3.1. Determination of reducing sugars

The amount of reducing sugars in the media was determined using the methods of Noeling and Bernfeld (1948) and Carneiro et al. (2004), with some modifications. First, all the media were hydrolyzed. The sugar

extract (1 mL) was added to 50 μL of concentrated hydrochloric acid (analytical grade) in a glass test tube. The tubes were shaken using a vortex for homogenization and placed in boiling water (approximately 100 °C) for 20 min. Distilled water (3000 μL) was then added to the tube.

After hydrolysis, 200 μL of the sugar extract was added to 200 μL of 3,5-dinitrosalicylic acid (DNS) solution in another test tube. The tubes were then placed in boiling water (approximately 100 °C) for 5 min and subsequently cooled to room temperature. After cooling, 1600 μL of distilled water was added to the same tube and stirred using a vortex mixer for homogenization. The absorbance was measured against a blank at a wavelength of $\lambda = 520$ nm using a spectrophotometer (SP-220, Biospectro, Brazil) in a glass cuvette (volume of 3 mL). The blank consisted of distilled water, which was subjected to the same procedure. The total soluble sugar content was expressed in g of glucose/100 g of extract.

2.4. Characterization of xanthan gum

2.4.1. Apparent viscosity

The apparent viscosity of commercial and laboratory-produced xanthan gum solutions was evaluated as a function of shear rate using a parallel plate geometry rheometer (Modular Compact Rheometer MCR 102e, Anton Paar, Austria) equipped with a 50 mm diameter plate. Solutions were prepared by magnetic stirring overnight (500 rpm) at room temperature, at a concentration of 0.5 % (w/v) XG. Rheological measurements were performed at four temperatures: 25 °C, 45 °C, 65 °C, and 85 °C and shear rates between 25 and 1000 s^{-1} . The results are presented as plots of apparent viscosity versus shear rate.

The apparent viscosity (η_{ap}) of non-Newtonian fluids was modeled using the Ostwald-de-Waele equation (Eq. 1):

$$\eta_{ap} = K(\dot{\gamma})^{n-1} \quad (1)$$

Where " η_{ap} " is the apparent viscosity (Pa.s), " K " is the consistency index (Pa.s), " $\dot{\gamma}$ " is the shear rate (s^{-1}) and " n " is the flow behavior index (dimensionless). The parameters K , n , and the coefficient of determination (R^2) were obtained by nonlinear regression analysis, enabling quantitative comparison of the rheological behavior across different temperatures and sample types (Gomes et al., 2015).

2.4.2. Intrinsic viscosity - XG average molar mass

The intrinsic viscosity (η) of the polymeric solutions was determined using the Mark-Houwink equation (Eq. 2), which is related to the average molecular weight of the polymer (Soleymanpour et al., 2018).

$$\eta = K(M)^a \quad (2)$$

Where " K " and " a " are constants and depend on the polymer, solvent and temperature used in the viscometry test. In this case, $a = 1.140$ and $K = 0.0000017$ (dl/mg) according to the American Polymer Standards Corporation (1983) at 25 °C. Sodium chloride (NaCl) 0.1 M was used as the solvent.

The intrinsic viscosity (η) was obtained by extrapolating the inherent viscosities (η_{iner}) and specific reduced (η_{espred}) solutions diluted to zero concentration from the intercept in the graph versus xanthan concentration. The molar mass was calculated using the Mark-Houwink equation (Eq. 2) (Soleymanpour et al., 2018).

The samples used for the determination of molar masses by the viscosity method were prepared using 0.1 % (w/v) xanthan gum (commercial and laboratory-produced). The solution was prepared via magnetic stirring overnight at 500 rpm at room temperature. The 0.1 % solution was diluted to reach the concentrations 0.05, 0.025, and 0.0125 %.

The intrinsic viscosity was measured using a Brookfield viscometer (model DV-II+ Pro, Brookfield, USA) with a Spindle S-00 at room temperature coupled to a water bath. For this viscometer model, 25 mL of an aqueous xanthan gum solution was required. Each sample was

transferred to the equipment and a variable rotation speed (50 rpm) was applied to the metallic axes immersed in the liquid.

2.4.3. Thermogravimetric analysis (TGA)

Thermogravimetric curves were used to determine the temperature range that represented the initial and final degradation temperatures of each thermic event. The first derivative of the TGA, DTG curve, was used to determine the rate of mass loss, Tmax (maximum thermal decomposition temperature), of each thermic event, and to identify specific degradation steps and stability of the thermal degradation of the samples. Approximately 2 mg of polymers were taken on an alumina pan and heated from 50 °C to 800 °C with a heating rate of 20 °C/min under continuous nitrogen gas flow of 50 cm^3/min in a DTG-60 H (Shimadzu, Japan) (Sahora et al., 2022).

2.4.4. Differential scanning calorimeter (DSC)

Differential scanning calorimetry was used to determine the thermal behavior and thermal decomposition profile of the xanthan gum. Analysis was performed using a Shimadzu DSC-60 Plus instrument (Shimadzu Corporation, Kyoto, Japan). Approximately 2 mg of xanthan gum was weighed and placed in an aluminum pan under a nitrogen atmosphere (50 mL/min) at a heating rate of 20 °C/min and a temperature range of 25–600 °C (Jin et al., 2022).

2.4.5. Fourier Transform Infrared Spectroscopy (FTIR)

The specific chemical groups and interactions between the polymer chains were characterized using FTIR spectroscopy (Brunchi et al., 2016; Chen et al., 2022a). Samples were measured at 400–4000 cm^{-1} with 64 scans (128 scans for background) and a resolution of 4 cm^{-1} using an FT-IR spectrometer (PerkinElmer Spectrum 3) in the ATR mode (Chen et al., 2022a). Linear baselines were applied to each absorbance peak, and three characteristic absorbance bands were chosen for analysis (Beck et al., 2018; Ponzeccchi et al., 2024): the 1027 cm^{-1} band, which is a typical band of polysaccharides (stretching of the C-C and C-O molecular groups), the 1656 cm^{-1} band, which is attributed to the stretching vibrations of the carboxylate anion, and the 3416 cm^{-1} stretching vibration of H-bonded -OH (Gomes et al., 2015).

The areas under the selected FTIR peaks were estimated using the trapz function in Matlab (version R2023b). Ratios were calculated by dividing the area of one peak by another to be able to compare chemical characteristics semi-quantitatively between spectra.

2.4.6. Elemental analysis of XGs

Elemental analysis (percentage of carbon, hydrogen, and nitrogen) of the samples was performed using a CHNS analyzer (Eurovector EA 5000). The samples consisted of between 0.5 and 1 mg of solid sample, which was analyzed under the following conditions: drag gas He (5.0–110 kPa), purge flow rate (80 mL/min), oxygen (15 mL), oxygen pressure delta (35 kPa), oxidation time of 6.6 s, sample delay of 5 s, running time of 400 s, reactor temperature of 980 °C, and oven temperature of 100 °C. Quantification was performed using a sulfanilamide standard. The oxygen content was estimated by subtracting the sum of the carbon, hydrogen, and nitrogen percentages from 100 % (Musule et al., 2022).

3. Results and discussion

3.1. Production of xanthan gum in the laboratory

The focus of this research was to use agro-industry residues that are readily available in Minas Gerais (Brazil) as a low-cost alternative substrate. This approach aims to repurpose waste, reduce biopolymer production costs (see Supplementary Material), and, consequently, reduce the costs of biodegradable food packaging.

According to García-Ochoa et al. (2000), the concentration of the carbon source affects xanthan yield, and a concentration of 2–4 % is

preferred. Furthermore, according to Gomes et al. (2015), a lower content of sugarcane bagasse (2 %) resulted in low production of xanthan due to the lack of carbon broth. According to the authors, a higher content of sucrose (6, 8 %, and 10 %) could limit bacterial development (inhibition by substrate) due to excess carbon. Based on this information, preliminary tests were conducted using sugarcane bagasse at various concentrations. The objective of this part of the study was to determine the appropriate concentration of the substrate for xanthan production.

During the fermentation process, reducing sugars (DNS method) were determined to analyze the total sugar content in the medium. As shown in Fig. 1, the medium containing 2 % (w/v) sugarcane bagasse extract exhibited a noticeable reduction in glucose levels by the second day (0.62 g glucose/100 mL medium), indicating reduced fermentative activity. Even with 2 % (v/v) broth supplementation, the increase to 0.71 g/100 mL was not enough to support sustained fermentation. Based on the results, it was proposed to triple the sugar concentration in the medium, establishing the concentration of sugarcane bagasse extract as 6 % (w/v). The total sugar content reached 1.78 g glucose/100 mL medium by the 5th day. Fig. 1 shows the total sugar concentration of each fermentation medium.

According to Rosalam and England (2005), different substrates do not influence the primary backbone structure of xanthan gum, but do affect the structure of side-chains, the molecular mass, and yield. Additionally, Gomes et al. (2015), reported that when different residues and fermentation media are used, the bacteria develop a distinct metabolic route, which leads to a biopolymer with a different molar mass, while maintaining the essential polymeric properties.

The mean yield obtained for xanthan gum with 6 % (w/v) sugarcane bagasse extract in our experiment was 2.07 g/L. This value is consistent with the literature. Gomes et al. (2015) reported xanthan gum production yields ranging from 0.54 g/L to 8.44 g/L. The maximum yield was achieved when the sugarcane bagasse concentration was 4 % (w/v). The second-highest yield was achieved at a substrate concentration of 6 % (w/v), resulting in a yield of approximately 4 g/L. The low production obtained with 2 % (w/v) can be attributed to the lack of carbon in the culture broth, as observed in the results developed in their research. The sugar content in sugarcane depends on its degree of maturation and is influenced by factors such as soil and climate conditions, in addition to specific sugarcane variety.

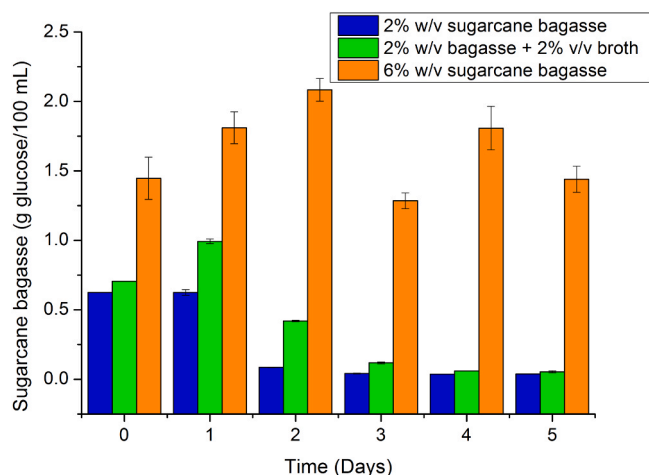


Fig. 1. Glucose concentration over 5 days in different fermentation media. Blue bars represent 2 % w/v sugarcane bagasse; green bars represent 2 % w/v bagasse + 2 % v/v broth; orange bars represent 6 % w/v sugarcane bagasse. Glucose concentration increased over time in all conditions, with the 6 % w/v bagasse consistently showing the highest values. Error bars indicate standard deviation.

3.2. Characterization of xanthan gum

3.2.1. Thermal properties (TGA, DTG and DSC)

The TGA/DTG curves of xanthan gum produced in the laboratory and the commercial gum from Synth, are shown in Fig. 2 and were used to determine the temperature ranges corresponding to the percentages of the two thermogravimetric curves, dehydration and decomposition of the organic material (Table 1). It is possible to observe that both xanthan gums, commercial from Synth (XGcom), and those produced in the laboratory using sugarcane bagasse (XGlab), present similar thermal events and stability when compared to each other.

The first thermal event corresponds to the dehydration of the polymer, which involves the loss of surface and intrinsic water associated with the saccharide structure of XG through hydrogen bonds (Raschip et al., 2013; Soleymanpour et al., 2018). This occurred at temperatures ranging from 30° to 185° C, with a temperature of maximum mass loss at 65° C (16.27 %) for XGcom, and 30° to 160° C, with Tmax at 52° C (16.27 %) for XGlab. The second mass loss corresponds to the degradation of the polymer, occurring between 205° and 380° C (47.70 %) for XGcom and between 210° and 370° C (47.18 %) for XGlab. This stage involves depolymerization and pyrolysis, leading to the structural decomposition of the polysaccharide chain (Raschip et al., 2013). The maximum decomposition temperature is higher for XGcom (300° C), however, laboratory-produced xanthan gum has a thermal decomposition temperature (290° C) close to the commercial xanthan. This suggests that the biopolymer produced under different operating conditions using agro-industry residue processes comparable thermal stability to the commercial gum.

These data are consistent with those found by Gomes et al. (2015). The dehydration of the polymer produced from sugarcane bagasse occurs between 29.6° to 181.1° C and the degradation occurs between 222.1° to 353.7° C. Zohuriaan and Shokrolahi (2004) studied the thermal curves of xanthan gum (Aldrich) and found that the dehydration of the polymer occurs between 30° to 91.3° C (Tmax at 74.6° C and 8.7 % of mass loss). The second thermic event occurred between 251.5° to 330.3° C (Tmax at 290.8° C and 41.6 %), and a final residue of 12 %. Raschip et al. (2013) reported two thermic events for xanthan gum, with the maximum decomposition temperature occurring in 267° C, accompanied by a 53 % of mass loss.

Xanthan gum produced using an alternative substrate exhibited a higher amount of pyrolysis residue (17.35 %) suggesting the presence of more inorganic materials compared to the commercial biopolymer, due to the substrate and the simple operational conditions used, without prior purification. However, this outcome aligns with the aim of the project, which is to develop a biopolymer through a simple and low-cost process that utilizes agro-industrial waste, eliminating the need for purification steps and reducing production costs, with the goal of manufacturing biodegradable food packaging.

The DSC curves of xanthan gum produced in the laboratory and the commercial gum from Synth, are shown in Fig. 3. It can be observed that they have a similar curve behavior, presenting two thermic events. The first occurred between 30 and 180° C, showing an endothermic peak around 65° C, attributed to the polymer dehydration. The second event occurred between 210° to 370° C, showing an exothermic peak around 300° C, that is related to polysaccharide degradation (Da Silva et al., 2018). This thermal event observed on the DSC curve is reflected on the TGA/DTG curves.

Similar results for XG from Sigma were found by Da Silva et al. (2018). Maia et al. (2012) reported both endothermic and exothermic peaks for commercial xanthan gum. The endothermic peak occurred below 100° C due to polymer dehydration, while the exothermic peak appeared at 275° C, corresponding to polysaccharide degradation. According to Zohuriaan and Shokrolahi (2004), generally, the degradation process is exothermic, as the formed products (CO, CO₂, CH₄) are more stable than the polysaccharide chain.

The thermal stability is particularly relevant for future application in

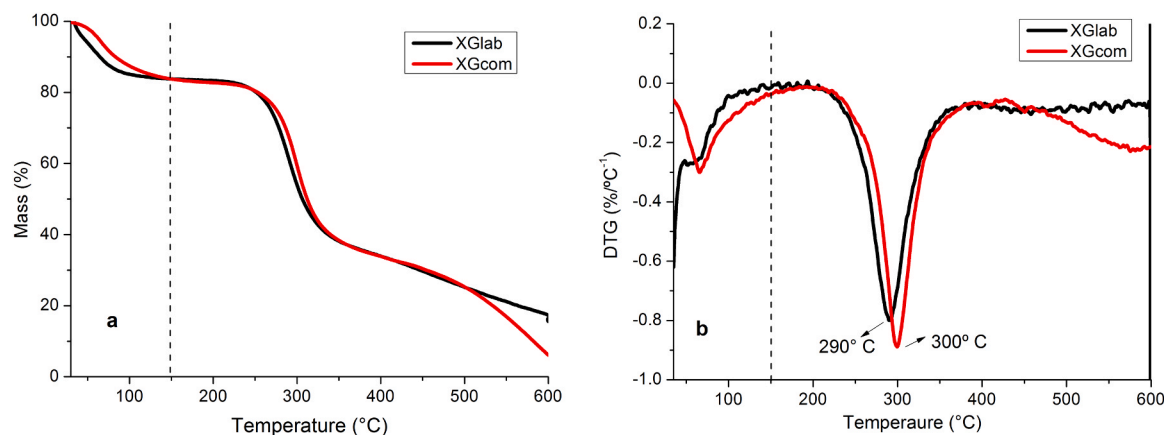


Fig. 2. Thermogravimetric (TG) and derivative thermogravimetric (DTG) curves of xanthan gum samples. (a) TG curves showing mass loss (%) as a function of temperature. (b) DTG curves showing decomposition peaks at approximately 290 °C for the laboratory-produced sample (XGlab) and 300 °C for the commercial sample (XGcom), indicating differences in thermal stability. The black line corresponds to commercial xanthan gum (Synth), while the red line represents the sample produced in the laboratory using sugarcane bagasse.

Table 1

Thermal events observed in commercial and laboratory xanthan gum samples, including temperature ranges (°C), maximum temperature (Tmax - °C), and weight loss (%).

Samples	Thermal event	Temperature range (°C)	Tmax (°C)	Weight loss (%)
XGcom	Dehydration	30–185	65	16.27
	Degradation	205–380	300	47.70
	Residue (600° C)	-	-	6.15
XGlab	Dehydration	30–160	52	16.27
	Degradation	210–370	290	47.18
	Residue (600° C)	-	-	17.35

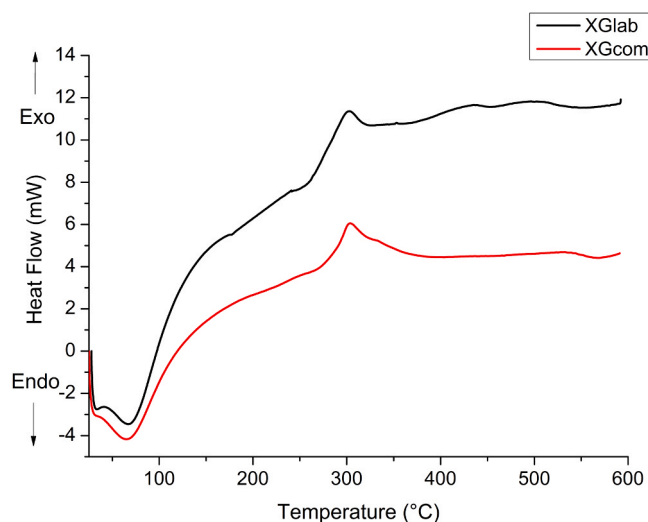


Fig. 3. Differential scanning calorimetry (DSC) curves of xanthan gum samples: XGlab (black line, produced in the laboratory using sugarcane bagasse) and XGcom (red line, commercial sample from Synth). Both samples exhibit similar thermal profiles, with two distinct events: an endothermic peak around 65 °C, related to polymer dehydration, and an exothermic peak near 300 °C, associated with polysaccharide degradation. These transitions reflect comparable structural characteristics and are consistent with the thermal behavior observed in the TGA/DTG analysis.

biodegradable packaging, as polymeric materials must withstand

processing conditions that often involve higher temperatures. According to [Elella et al. \(2021\)](#), the ordered helical structure of XG contributes to its resistance against thermal degradation, making it suitable for use in formulations that require heat treatment. Ensuring thermal stability during processing is essential to maintain the integrity and performance of the final packaging material.

In this context, it is worth noting that many traditional organic polymers begin to decompose at temperatures ranging from 150 °C to 300 °C, which can lead to degradation reactions during processing ([Coutinho et al., 2003](#)). Specifically, low-density polyethylene (LDPE), the main polymer used in flexible food packaging, degrades between 110 °C and 150 °C ([Coutinho et al., 2003](#)). According to [Cáceres and Canevarollo \(2009\)](#), this behavior is associated with chain scission distribution mechanisms occurring above 220 °C, depending on the initial molar mass and extrusion conditions.

The xanthan gum produced from sugarcane bagasse demonstrated thermal stability within a range that is compatible with these processing requirements, reinforcing its potential as a sustainable alternative to conventional polymers in biodegradable packaging applications.

3.2.2. Viscosity (apparent and intrinsic)

The apparent viscosities of both xanthan gum samples, commercial and laboratory-produced, as a function of shear rate (25–1000 s⁻¹), are shown in [Fig. 4](#). As the shear rate increases, viscosity decreases, confirming the non-Newtonian nature of both solutions, typical of viscoelastic fluids. This shear-thinning behavior is a well-known feature of xanthan gum and reflects its pseudoplastic character. The reduction in viscosity arises from the alignment of xanthan chains in the direction of flow, which reduces intermolecular interactions and entanglements, regardless of whether the molecules adopt a helical or random coil conformation. Pseudoplastic flow is therefore observed independently of the conformational state ([Whitcomb and Macosko, 1978](#); [Rochefort and Middleman, 1987](#); [Pelletier et al., 2001](#); [Choppe et al., 2010](#); [Fitzpatrick et al., 2013](#)).

The influence of temperature on viscosity was also evident for both xanthan gum samples, which exhibited a progressive decrease in apparent viscosity as temperature increased. This behavior is consistent with the temperature sensitivity of xanthan molecular structure, since higher temperatures promote disruption of the ordered helical conformation and favor the random coil state, thereby reducing chain entanglement and intermolecular associations ([Pelletier et al., 2001](#); [Choppe et al., 2010](#)).

The apparent viscosity at 25 °C and 25 s⁻¹ for the commercial and laboratory-produced xanthan gums was 311 and 135 mPa·s, respectively. [Gomes et al. \(2015\)](#) used sugarcane bagasse (6 % w/v) as a

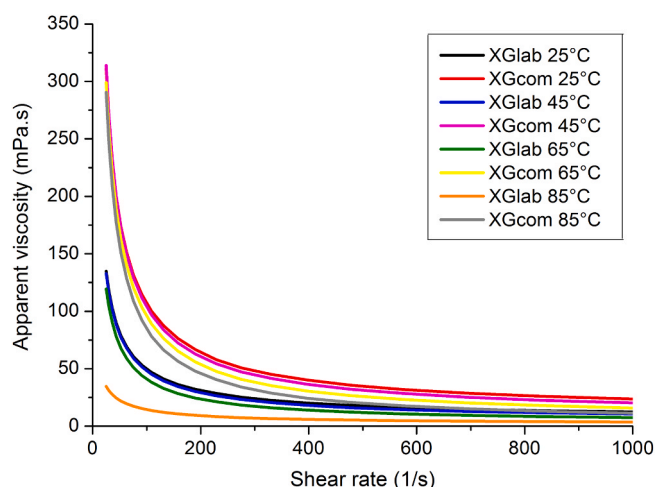


Fig. 4. Apparent viscosity profiles of XGlab and XGcom samples at different temperatures (25°C, 45°C, 65°C, and 85°C) as a function of shear rate (0–1000 s⁻¹). All samples exhibit shear-thinning behavior, with viscosity decreasing as shear rate increases. The curves are color-coded to indicate sample type and temperature: XGlab (black, blue, green, orange) and XGcom (red, magenta, yellow, gray).

carbon source and reported a viscosity of approximately 75 mPa·s, while [Brandão et al. \(2008\)](#) found around 30 mPa·s for xanthan gum produced from sugarcane broth. [Da Silva et al. \(2018\)](#) used coconut shell (CS) and cocoa husks (CH) as substrates to produce xanthan gum with two different *X. campestris* strains. The viscosities obtained were 134.20 and 112.06 mPa·s for CS, and 120.51 and 72.46 mPa·s for CH, respectively. These values are comparable to that of commercial xanthan gum (Sigma), which showed a viscosity of 295.85 mPa·s. Rheological characteristics, such as apparent viscosity and pseudoplastic behavior, are influenced by factors including chemical composition, biopolymer structure (e.g., chain rearrangement and molecular interactions), average molecular weight, and gum concentration ([Nery et al., 2008](#)).

To evaluate the pseudoplastic behavior of both samples at 0.5 % (w/v), the experimental data were fitted to the Ostwald-de-Waele power law model. The regression yielded high correlation coefficients (R^2 ranging from 0.997 to 0.999), confirming the suitability of the model for describing the flow behavior of xanthan gum solutions, as presented in [Table 2](#). The consistency index (K) reflects the resistance of the fluid to flow, while the flow behavior index (n) was consistently below unity ($n < 1$), indicating pronounced pseudoplastic behavior ([Nery et al., 2008](#); [Gomes et al., 2015](#)). Values were obtained from single measurements.

The molecular weight in Daltons, estimated from intrinsic viscosity, was approximately 1018 kDa (g/mol) for commercial xanthan (Sigma) and 1238 kDa for the xanthan produced in the laboratory. These results are consistent with the literature, [Masuelli \(2014\)](#) determined the molecular weight of xanthan gum (Parafarm) and found 1675 kDa. Furthermore, 2000 kDa was calculated by [Cuvelier and Launay \(1986\)](#).

Table 2

Flow behavior index (n , dimensionless), consistency index (K , Pa·s) and coefficient of determination (R^2 , dimensionless) for commercial (XGcom) and laboratory (XGlab) xanthan gum samples at 0.5 % (w/v), obtained using the Ostwald-de-Waele model at different temperatures.

T (°C)	XGlab			XGcom		
	n	K	R^2	n	K	R^2
25	0.343	1052.79	0.997	0.290	2893.73	0.997
45	0.301	1217.90	0.998	0.247	3383.28	0.998
65	0.236	1382.85	0.999	0.192	3966.34	0.999
85	0.370	261.62	0.999	0.112	5058.88	0.999

[Soleymanpour et al. \(2018\)](#) found 2230 kDa for xanthan gum produced from broomcorn stem, with intrinsic viscosity estimated using the Mark-Howink equation. [Gomes et al. \(2015\)](#) used a different methodology (HPLC) to obtain the average molecular weight of XG and found for 6 % (w/v) of sugarcane bagasse 338,650 Da and for Sigma, 331,780 Da. Despite variations in the methodologies and sources of xanthan, the values obtained in this study are within the range reported in the literature and confirm the consistency among different samples.

According to [García-Ochoa et al. \(2000\)](#), the chemical composition, fermentation medium, strain of *Xanthomonas campestris*, and operational conditions can influence both the yield and molecular weight of xanthan gum. A higher molar mass is directly associated with increased intrinsic viscosity. [Oliveira et al. \(2023\)](#) also emphasize that, for natural polysaccharides, the composition and chain length distribution significantly affect rheological behavior. In this context, elevated molecular weights are particularly important for the development of biopolymer films with desirable mechanical and barrier properties.

The thermal sensitivity of xanthan gum viscosity reinforces its relevance in polymer blending and film-forming processes, where temperature and flow behavior are key parameters. Its high viscosity and pronounced pseudoplasticity, combined with favorable rheological properties, contribute directly to film formation. Moreover, its ability to disperse in both cold and hot water, while maintaining viscosity across a wide range of temperatures and pH levels, ensures compatibility with various fabrication conditions ([Elella et al., 2021](#)). These attributes are essential for the development of biodegradable food packaging, where processing stability and functional performance are critical.

3.2.3. Chemical characterization (elemental analysis)

Elemental analysis was performed to evaluate the composition and similarity between XGlab and XGcom, and the results are summarized in [Table 3](#). The commercial xanthan gum exhibited a predominantly polysaccharide composition, consisting mainly of carbon, hydrogen and oxygen. XGlab presented similar carbon and hydrogen content, however, nitrogen was detected, leading to a reduction in the oxygen percentage. This nitrogen content is due to the use of urea as a nitrogen source in the biosynthesis process, which is an essential nutrient ([García-Ochoa et al., 2000](#)). Additionally, traces of impurities or residues from the culture medium, such as proteins, yeast extract, or peptone, may also contribute to this result. For XGlab, only two measurements were conducted due to experimental constraints, and therefore standard deviations could not be calculated. The values presented represent the mean of these two measurements.

According to [García-Ochoa et al. \(2000\)](#), the primary structure of xanthan gum consists of repeated pentasaccharide units, which are mainly composed of carbon, hydrogen and oxygen. According to [Elella et al. \(2017\)](#), traces of nitrogen in analysis of XG might be because of the presence of impurities such as proteins from XG production during the fermentation process. The authors found 37.77 %, 6.37 %, 55.23 % and 0.63 % for C, H, O and N, respectively, for xanthan gum. The similarity in carbon and hydrogen content suggests that the primary structure of the biopolymer remains consistent with literature reports, even when different substrates and fermentation conditions are used.

Understanding the elemental composition and structural consistency of xanthan gum is essential, as these characteristics directly influence its behavior when incorporated into polymeric matrices. [Chen et al. \(2022a\)](#) found that the incorporation of XG into PVA films influenced

Table 3

Elemental analysis of commercial (XGcom) and laboratory (XGlab) xanthan gum samples, showing the contents of carbon, hydrogen, nitrogen, and sulfur.

Sample	C (%)	H (%)	N (%)	O (%)
XGlab	37.69*	5.25*	3.02*	54.04*
XGcom	37.42 ± 1.04	5.08 ± 0.19	-	57.50 ± 0.84

* Mean values

the degradation process, making the polymer blend more susceptible to microbial degradation when exposed to water and soil environments.

3.2.4. Chemical characterization (FTIR)

The specific chemical groups of the two xanthan gums were characterized by ATR-FTIR. The experimental FTIR spectra of XGcom and XGlab are shown in Fig. 5. The spectra confirm that xanthan gum is a polymer rich in hydroxyl, acetyl and carboxylic acid groups (Zhang et al., 2019).

Xanthan gum produced in the laboratory and the standard polymer has absorption bands at approximately 3350 cm^{-1} and 2900 cm^{-1} due to respectively hydrogen bonded -OH and C-H stretching. Absorptions at 1600 cm^{-1} and 1370 cm^{-1} are related to asymmetric and symmetric stretching vibrations of the carboxylate anion. Furthermore, absorptions at approximately 1027 cm^{-1} , can be associated with C-C and C-OH (Kačuráková et al., 1999).

We chose to focus on the relative intensities of the three bands at 3351 , 1602 and 1016 cm^{-1} (Figs. 6a and 6b). As expected from literature (Gomes et al. (2015) and Soleymannpour et al. (2018)) the ratios between the XGcom and XGlab absorbance peaks are similar, indicating that the biosynthesis process based on an agro-industrial waste can produce xanthan gum with a structure similar to gum based on sucrose.

The presence of hydroxyl and carboxylate groups in both samples confirms the chemical functionality required for film formation and polymer blending. These functional groups enable strong hydrogen bonding interactions, which are essential for compatibility with other polymers (Brunchi et al., 2016). The interactions are particularly relevant in the development of composite films, where xanthan gum can reduce water solubility, improve mechanical strength and barrier properties (Chen et al., 2022; Zhang et al., 2019).

The chemical composition and structural consistency observed in XGlab reinforce its suitability for use in biodegradable packaging systems. The presence of functional groups that promote polymer interactions supports its integration into environmentally friendly materials. By utilizing agro-industrial residues in its production, this biopolymer offers a sustainable alternative to conventional plastics, aligning with current demands for low-impact packaging solutions.

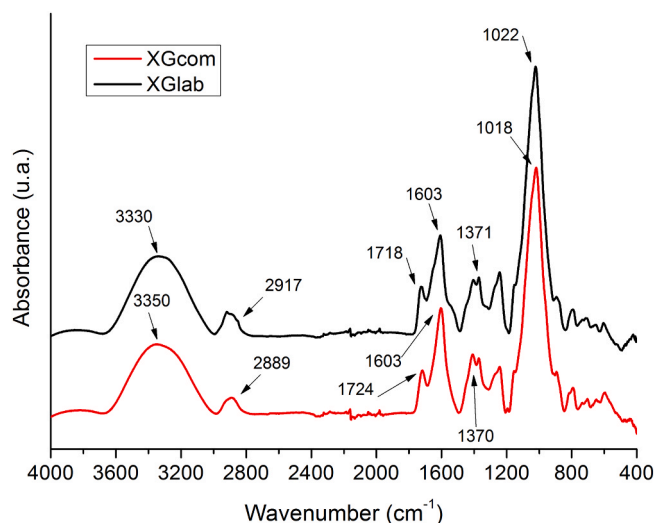


Fig. 5. FTIR spectra of XGlab (black line) and XGcom (red line) samples in the range of $4000\text{--}400\text{ cm}^{-1}$. Characteristic absorption bands are observed for both samples, including O-H stretching ($\sim 3330\text{--}3350\text{ cm}^{-1}$), C-H stretching ($\sim 2917\text{ cm}^{-1}$ for XGlab and $\sim 2889\text{ cm}^{-1}$ for XGcom), C=O stretching ($\sim 1718\text{ cm}^{-1}$ for XGlab and $\sim 1724\text{ cm}^{-1}$ for XGcom), and C-O-C vibrations ($\sim 1018\text{--}1022\text{ cm}^{-1}$). Differences in peak positions and intensities suggest structural and compositional variations between the samples.

4. Conclusion

Plastic pollution remains a critical environmental challenge due to the widespread use of petroleum-based polymers in food packaging, and the high cost of biopolymers has limited their adoption as sustainable alternatives. In this study, sugarcane bagasse, an abundant agro-industrial residue in Brazil, was successfully used as a low-cost substrate for xanthan gum production through a simplified fermentation process. The gum obtained exhibited physicochemical properties comparable to those of commercial xanthan, including pseudoplastic rheology, high molecular weight, and thermal stability suitable for processing. These findings demonstrate the feasibility of valorizing agro-industrial waste into functional biopolymers, contributing to circular resource use and reducing production costs. The structural and functional similarities between lab-produced and commercial xanthan gum reinforce its potential for incorporation into polymeric matrices, particularly in biodegradable food packaging films. Looking ahead, future research should focus on the development of composite films by blending xanthan gum with polyvinyl alcohol (PVA) and other biodegradable polymers, aiming to improve mechanical strength, barrier properties, and water resistance for practical applications in food packaging. Altogether, this work highlights sugarcane bagasse as a sustainable and cost-effective feedstock for xanthan gum production and supports its application in environmentally responsible packaging solutions, offering a viable alternative to conventional plastics.

CRediT authorship contribution statement

Marali Vilela Dias: Writing – review & editing, Methodology, Conceptualization. **Enio Nazaré de Oliveira Junior:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Maria Rita Meyer Ferraz da Costa:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Anand Ramesh Sanadi:** Writing – review & editing, Methodology. **Gustavo Henrique Denzin Tonoli:** Conceptualization. **Mario Guimarães Junior:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Lisbeth Garbrecht Thygesen:** Writing – review & editing, Methodology, Data curation.

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Ethical considerations

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Maria Rita Meyer Ferraz da Costa reports financial support was provided by Coordination for the improvement of Higher Education Personnel. Maria Rita Meyer Ferraz da Costa reports financial support was provided by National Council for Scientific and Technological Development.

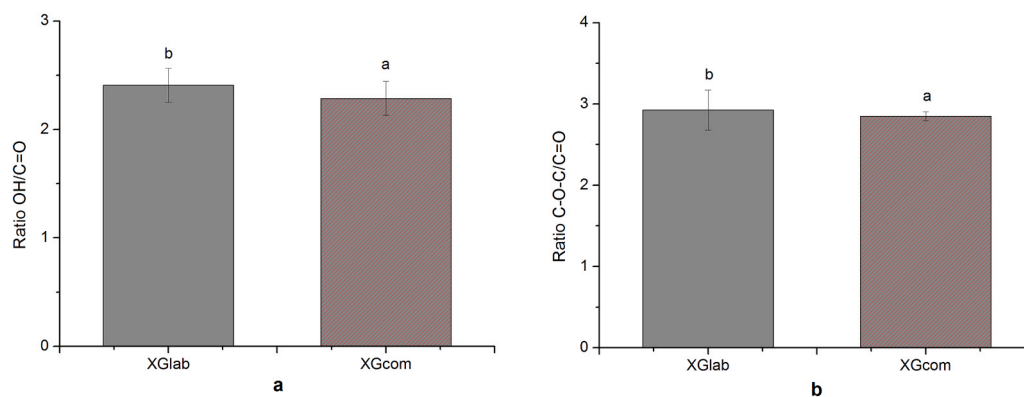


Fig. 6. Quantitative comparison of FTIR peak ratios for XGlab and XGcom samples. (a) Ratio of hydroxyl to carbonyl groups (OH/C=O), showing a higher value for XGlab, indicating greater hydroxyl content. (b) Ratio of ether to carbonyl groups (C–O–C/C=O), also higher in XGlab, suggesting a more branched or crosslinked structure. Bars labeled with different letters (a, b) indicate statistically significant differences ($p < 0.05$).

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fbp.2026.01.008](https://doi.org/10.1016/j.fbp.2026.01.008).

Data availability statement

All data generated or analyzed during this study are included in this published article [and its supplementary information files].

References

- Beck, G., Thybring, E.E., Thygesen, L.G., 2018. Brown-rot fungal degradation and deacetylation of acetylated wood. *Int. Biodeterior. Biodegrad.* 135, 62–70. <https://doi.org/10.1016/j.ibiod.2018.09.009>.
- Brandão, L.V., Nery, T.B.R., Machado, B.A.S., et al., 2008. Produção de goma xantana obtida a partir do caldo de cana. *Food Sci. Technol.* 28, 217–222. <https://doi.org/10.1590/S0101-20612008000500033>.
- Brandão, L.V., Assis, D.J., López, J.A., et al., 2013. Bioconversion from crude glycerin by *Xanthomonas campestris* 2103: xanthan production and characterization. *Braz. J. Chem. Eng.* 30, 737–746. <https://doi.org/10.1590/S0104-66322013000400006>.
- Brunchi, C.-E., Bercea, M., Morariu, S., et al., 2016. Investigations on the interactions between xanthan gum and poly(vinyl alcohol) in solid state and aqueous solutions. *Eur. Polym. J.* 84, 161–172. <https://doi.org/10.1016/j.eurpolymj.2016.09.006>.
- Cáceres, C.A., Canevarollo, S.V., 2009. Degradação do polipropileno durante a extrusão e a geração de compostos orgânicos voláteis. *Polímeros* 19, 79–84. <https://doi.org/10.1590/S0104-14282009000100017>.
- Carneiro, C.N.B., Isejima, E.M., Samuels, R.I., et al., 2004. Sucrose hydrolases from the midgut of the sugarcane stalk borer *Diatraea saccharalis*. *J. Insect Physiol.* 50, 1093–1101. <https://doi.org/10.1016/j.jinsphys.2004.09.011>.
- Chen, J., Zheng, M., Tan, K.B., et al., 2022a. Development of xanthan gum/hydroxypropyl methyl cellulose composite films incorporating tea polyphenol and its application on fresh-cut green bell peppers preservation. *Int. J. Biol. Macromol.* 211, 198–206. <https://doi.org/10.1016/j.ijbiomac.2022.05.043>.
- Chen, J., Zheng, M., Tan, K.B., et al., 2022b. Polyvinyl alcohol/xanthan gum composite film with excellent food packaging, storage and biodegradation capability as potential environmentally-friendly alternative to commercial plastic bag. *Int. J. Biol. Macromol.* 212, 402–411. <https://doi.org/10.1016/j.ijbiomac.2022.05.119>.
- Choppe, E., Puaud, F., Nicolai, T., Benyahia, L., 2010. Rheology of xanthan solutions as a function of temperature, concentration and ionic strength. *Carbohydr. Polym.* 82, 1228–1235. <https://doi.org/10.1016/j.carbpol.2010.06.056>.
- Coutinho, F., Mello, I.L., Santa Maria, L.C., 2003. Polietileno: principais tipos, propriedades e aplicações. *Polímeros* 13, 1–13. <https://doi.org/10.1590/S0104-14282003000100005>.
- Cuvelier, G., Launay, B., 1986. Concentration regimes in xanthan gum solutions deduced from flow and viscoelastic properties. *Carbohydr. Polym.* 6, 321–333. [https://doi.org/10.1016/0144-8617\(86\)90023-8](https://doi.org/10.1016/0144-8617(86)90023-8).
- Da Silva, J.A., Cardoso, L.G., Assis, D.J., et al., 2018. Xanthan gum production by *Xanthomonas campestris* pv. *campestris* IBSBF 1866 and 1867 from lignocellulosic agroindustrial wastes. *Appl. Biochem. Biotechnol.* 186, 750–763. <https://doi.org/10.1007/s12010-018-2765-8>.
- Dos Santos, F.P., Oliveira, J.R.A.M., Nunes, T.P., et al., 2016. Bioconversion of agro-industrial wastes into xanthan gum. *Chem. Eng. Trans.* 49, 145–150. <https://doi.org/10.3303/CET1649025>.
- Elella, M.H.A., Mohamed, R.R., ElHafeez, E.A., et al., 2017. Synthesis of novel biodegradable antibacterial grafted xanthan gum. *Carbohydr. Polym.* 173, 305–311. <https://doi.org/10.1016/j.carbpol.2017.05.058>.
- Elella, M.H.A., Goda, E.S., Galb-Allah, M.A., et al., 2021. Xanthan gum-derived materials for applications in environment and eco-friendly materials: a review. *J. Environ. Chem. Eng.* 9, 104702. <https://doi.org/10.1016/j.jece.2020.104702>.
- Fitzpatrick, P., Meadows, J., Ratcliffe, I., Williams, P.A., 2013. Control of the properties of xanthan/glucuronan mixed gels by varying xanthan fine structure. *Carbohydr. Polym.* 92, 1018–1025. <https://doi.org/10.1016/j.carbpol.2012.10.049>.
- García-Ochoa, F., Santos, V.E., Casas, J.A., et al., 2000. Xanthan gum: production, recovery, and properties. *Biotechnol. Adv.* 18, 549–579. [https://doi.org/10.1016/S0734-9750\(00\)00050-1](https://doi.org/10.1016/S0734-9750(00)00050-1).
- Gomes, G.V.P., Assis, D.J., Silva, J.B.A., et al., 2015. Obtaining xanthan gum impregnated with cellulose microfibrils derived from sugarcane bagasse. *Mater. Today Proc.* 2, 389–398. <https://doi.org/10.1016/j.matpr.2015.04.042>.
- Hadi, A., Nawab, A., Alam, F., et al., 2022. Sustainable food packaging films based on alginate and Aloe vera. *Polym. Eng. Sci.* 62, 2111–2118. <https://doi.org/10.1002/pen.25992>.
- Hadidi, M., Jafarzadeh, S., Forough, M., et al., 2022. Plant protein-based food packaging films: recent advances in fabrication, characterization, and applications. *Trends Food Sci. Technol.* 120, 154–173. <https://doi.org/10.1016/j.tifs.2022.01.013>.
- Jayakumar, A., Radoor, S., Kim, J.T., et al., 2022. Recent innovations in bionanocomposites-based food packaging films – a comprehensive review. *Food Packag. Shelf Life* 33, 100877. <https://doi.org/10.1016/j.fpsl.2022.100877>.
- Jin, Q., Yu, H., Zhang, Y., et al., 2022. Influence of urea on polarization property and mechanical properties of polyvinyl alcohol films for the polarizers. *J. Polym. Res.* 29 (11). <https://doi.org/10.1007/s10965-022-03321-x>.
- Kačuráková, M., Wellner, N., Ebringerová, A., et al., 1999. Characterisation of xylan-type polysaccharides and associated cell wall components by FT-IR and FT-Raman spectroscopies. *Food Hydrocoll.* 13, 35–41. [https://doi.org/10.1016/S0268-005X\(98\)00067-8](https://doi.org/10.1016/S0268-005X(98)00067-8).
- Khalid, S., Zahid, M., Chaudhary, K., et al., 2024. Unveiling the emerging trends of egg components-based biodegradable food packaging development: a comprehensive review. *Compr. Rev. Food Sci. Food Saf.* 23 (5). <https://doi.org/10.1111/1541-4337.13433>.
- Li, H., Xing, R., Wang, Z., Li, G., 2025. Advancements in xanthan gum-based film and coating for food packaging. *Carbohydr. Polym.*, 123409. <https://doi.org/10.1016/j.carbpol.2025.123409>.
- Maia, A.M.S., Silva, H.V.M., Curti, P.S., et al., 2012. Study of the reaction of grafting acrylamide onto xanthan gum. *Carbohydr. Polym.* 90, 778–783. <https://doi.org/10.1016/j.carbpol.2012.05.059>.
- Murad, H.A., Abo-Elkhair, A.G., Azzaz, H.H., 2019. Production of xanthan gum from nontraditional substrates with perspective of the unique properties and wide industrial applications. *JSMC Microbiol* 1, 6.

- Musule, R., Pale-Ezquivel, I., Gordillo-Cruz, E., et al., 2022. A comparative analysis of different types of Mexican agroindustrial pellets using high-throughput instrumental techniques. *BioEnergy Res* 15, 1694–1710. <https://doi.org/10.1007/s12155-021-10370-5>.
- Masuell, M.A., 2014. Mark-Houwink Parameters for Aqueous-Soluble Polymers and Biopolymers at Various Temperatures. *J. Polym. Biopolym. Phys. Chem.* 2 (2), 37–43.
- Nery, T.B.R., Brandão, L.V., Esperidião, M.C.A., et al., 2008. Biossíntese de goma xantana a partir da fermentação de soro de leite: rendimento e viscosidade. *Quím. Nova* 31, 1937–1941. <https://doi.org/10.1590/S0100-40422008000800004>.
- Nery, T.B.R., Da Cruz, A.J.G., Druzian, J.I., 2013. Use of green coconut shells as an alternative substrate for the production of xanthan gum on different scales of fermentation. *Pol. ímeros* 23, 602–607. <https://doi.org/10.4322/polimeros.2013.094>.
- Noelting, G., Bernfeld, P., 1948. Sur les enzymes amylolytiques III. La β -amylase: dosage d'activité et contrôle de l'absence d' α -amylase. *Helv. Chim. Acta* 31, 286–290. <https://doi.org/10.1002/hlca.19480310149>.
- Nsengiyumva, E.M., Heitz, M.P., Alexandridis, P., 2024. Salt and temperature effects on xanthan gum polysaccharide in aqueous solutions. *Int. J. Mol. Sci.* 25, 490. <https://doi.org/10.3390/ijms25010490>.
- Oliveira, V.R.L., Coelho, V.V., Sérvulo, E.F.C., et al., 2023. Study of the molecular weight of pullulan produced by *Aureobasidium pullulans* from industrial waste. *Suppl. 1 Mater. Res.* 26. <https://doi.org/10.1590/1980-5373-MR-2023-0060>.
- Palaniraj, A., Jayaraman, V., 2011. Production, recovery and applications of xanthan gum by *Xanthomonas campestris*. *J. Food Eng.* 106, 1–12. <https://doi.org/10.1016/j.jfoodeng.2011.03.035>.
- Pelletier, E., Viebke, C., Meadows, J., Williams, P.A., 2001. A rheological study of the order–disorder conformational transition of xanthan gum. *Biopolymers* 59, 339–346. [https://doi.org/10.1002/1097-0282\(20011015\)59:5<339::AID-BIP1040>3.0.CO;2-3](https://doi.org/10.1002/1097-0282(20011015)59:5<339::AID-BIP1040>3.0.CO;2-3).
- Ponzecchi, A., Alfreiden, G., Fredriksson, M., et al., 2024. Localization and characterisation of brown rot in two types of acetylated wood. *Cellulose* 31, 1875–1890. <https://doi.org/10.1007/s10570-023-05680-0>.
- Rasch, I.E., Hitruc, G.E., Vasile, C., et al., 2013. Effect of the lignin type on the morphology and thermal properties of the xanthan/lignin hydrogels. *Int. J. Biol. Macromol.* 54, 230–237. <https://doi.org/10.1016/j.ijbiomac.2012.12.036>.
- Ratna, R., Mutia, M., Darwin, D., et al., 2024. Utilization of tofu liquid waste for the manufacture of bioplastic food packaging. *Case Stud. Chem. Environ. Eng.* 10, 100830. <https://doi.org/10.1016/j.csee.2024.100830>.
- Rocheffort, W.E., Middleman, S., 1987. Rheology of xanthan gum: salt, temperature, and strain effects in oscillatory and steady shear experiments. *J. Rheol.* 31, 337–369. <https://doi.org/10.1122/1.549953>.
- Rosalam, S., England, R., 2005. Review of xanthan gum production from unmodified starches by *Xanthomonas campestris* sp. *Enzym. Microb. Technol.* 39, 197–207. <https://doi.org/10.1016/j.enzmictec.2005.10.019>.
- Saroja, V., Khan, H., Raghuvanshi, S., et al., 2022. Development of polyvinyl alcohol-based antioxidant nanocomposite films with nanokaolin impregnated with polyphenols from pomegranate peel extract. *Food Packag. Shelf Life* 32, 100848. <https://doi.org/10.1016/j.fpsl.2022.100848>.
- Soleymannpour, Z., Nikzad, M., Talebnia, F., et al., 2018. Xanthan gum production from acid hydrolyzed broomcorn stem as a sole carbon source by *Xanthomonas campestris*. *3 Biotech* 8, 296. <https://doi.org/10.1007/s13205-018-1322-z>.
- Thongsomboon, W., Baimark, Y., Srihanam, P., 2023. Valorization of cellulose-based materials from agricultural waste: comparison between sugarcane bagasse and rice straw. *Polymers* 15, 3190. <https://doi.org/10.3390/polym15153190>.
- Ungureanu, N., Vlăduț, V., Biriș, S.-Ș., 2022. Sustainable valorization of waste and by-products from sugarcane processing. *Sustainability* 14, 11089. <https://doi.org/10.3390/su141711089>.
- Whitcomb, P.J., Macosko, C.W., 1978. Rheology of xanthan gum. *J. Rheol.* 22, 493–505. <https://doi.org/10.1122/1.549955>.
- Xu, R., Chaudhary, K., Altemimi, A.B., et al., 2025. Ultrasound-assisted extraction with deep eutectic solvents for valorization of agro-food waste: a sustainable and eco-friendly approach. *Ultrason. Sonochem.*, 107500. <https://doi.org/10.1016/j.ultsonch.2025.107500>.
- Zhang, Q., Hu, X.M., Wu, M.Y., et al., 2019. Synthesis and performance characterization of poly(vinyl alcohol)-xanthan gum composite hydrogel. *React. Funct. Polym.* 136, 34–43. <https://doi.org/10.1016/j.reactfunctpolym.2019.01.002>.
- Zohuriaan, M.J., Shokrolahi, F., 2004. Thermal studies on natural and modified gums. *Polym. Test.* 23, 575–579. <https://doi.org/10.1016/j.polymertesting.2003.11.001>.

Glossary

Term/Symbol: Definition

- XG:** Xanthan gum, a microbial polysaccharide with rheological properties
- XGlab:** Xanthan gum produced in the laboratory using sugarcane bagasse
- XGcom:** Commercial xanthan gum used as reference
- FTIR:** Fourier Transform Infrared Spectroscopy, used to identify functional groups
- TGA:** Thermogravimetric Analysis, measures mass loss with temperature increase
- DTG:** Derivative Thermogravimetry, shows rate of mass loss over temperature
- DSC:** Differential Scanning Calorimetry, identifies thermal events such as dehydration and degradation
- YM:** Yeast Malt medium, used for bacterial growth
- DNS:** 3,5-dinitrosalicylic acid, reagent for quantifying reducing sugars
- w/v:** Weight per volume, concentration unit
- v/v:** Volume per volume, concentration unit
- rpm:** Revolutions per minute, used for agitation speed
- μL:** Microliter, unit of volume
- NaCl:** Sodium chloride, used as solvent in viscosity tests
- MCR:** Modular Compact Rheometer, equipment for measuring viscosity
- ATR:** Attenuated Total Reflectance, technique used in FTIR spectroscopy